# Poach or Promote? Job Sorting and Gender Earnings Inequality across U.S. Industries

**by**

**Katariina Mueller-Gastell**
**Stanford University**

**CES 23-23** **April 2023**

The research program of the Center for Economic Studies (CES) produces a wide range of economic analyses to improve the statistical programs of the U.S. Census Bureau. Many of these analyses take the form of CES research papers. The papers have not undergone the review accorded Census Bureau publications and no endorsement should be inferred. Any opinions and conclusions expressed herein are those of the author(s) and do not represent the views of the U.S. Census Bureau. All results have been reviewed to ensure that no confidential information is disclosed. Republication in whole or part must be cleared with the authors.

To obtain information about the series, see www.census.gov/ces or contact Christopher Goetz, Editor, Discussion Papers, U.S. Census Bureau, Center for Economic Studies, 4600 Silver Hill Road, Washington, DC 20233, CES.Working.Papers@census.gov. To subscribe to the series, please click here.

**Abstract**

In this paper I outline the sociological theory that would predict that external labor markets – those in which more positions are filled with new hires rather from firm-internal promotions – heighten gender based discrimination and contribute to earnings inequality. I test this theory by treating industries as miniature labor markets within the US with varying levels of gender inequality and different hiring practices. Using high quality administrative data from 1985 to 2013, including detailed work histories from this period, I compare the earnings of alike men and women across industries with different levels of reliance on external markets at different times. I find that men experience greater unexplained earnings relative to women – unexplained in that it is not accounted for by work history or observable demographic characteristics – when a greater share of earnings increase events occur outside the firm.

* DISCLAIMER: Any views expressed are those of the authors and not those of the U.S. Census Bureau. The Census Bureau's Disclosure Review Board and Disclosure Avoidance Officers have reviewed this information product for unauthorized disclosure of confidential information and have approved the disclosure avoidance practices applied to this release. This research was performed at a Federal Statistical Research Data Center under FSRDC Project Number 2406. (CBDRB-FY22-P2406-R9600).

# Introduction

The so-called gender revolution has stalled, with few advances for women in the last couple of decades (Blau and Kahn, 2017; England, 2010; Gharehgozli and Atal, 2020). Part of this stagnation may be the result of a shifting employment landscape. Though interventions abound for boosting women's labor market prospects, be they formal, legal, or attitudinal (see e.g. Kalev et al., 2006), workplaces and employment relationships change rapidly, with more hiring occuring outside the firm (Bidwell et al., 2013).

Pay for the same work differs quite a bit across firms (Avent-Holt and Tomaskovic-Devey, 2019; Song et al., 2019). Thus, *where* you work is important for both current renumeration and for career trajectories. However, moving from a badly paying firm to a well paying firm may not be equally easy for all types of workers. Job seekers that are more likely to face discrimination

at the point of hire will have a more difficult time moving from below-average wage firms to above-average wage firms.

At the same time, churn and retention – whether workers move between firms or stay with the same employer – differ substantially by industry. In this chapter, I explore how differences in

---

*DISCLAIMER: Any views expressed are those of the authors and not those of the U.S. Census Bureau. The Census Bureau's Disclosure Review Board and Disclosure Avoidance Officers have reviewed this information product for unauthorized disclosure of confidential information and have approved the disclosure avoidance practices applied to this release. This research was performed at a Federal Statistical Research Data Center under FSRDC Project Number 2406. (CBDRB-FY22-P2406-R9600).

prevalent labor market strategies (i.e., whether careers progress internal to the firm or by moving between firms) interact with gender pay gaps. Does between-firm competition for workers affect the earnings of employees who are more likely to experience discrimination at the point of hire? Specifically, is the gender wage gap greater in industries that rely more heavily on external hiring?

In this chapter, I outline the sociological theory that would predict that external labor markets – those in which more positions are filled with new hires rather from firm-internal promotions – heighten gender based discrimination and contribute to earnings inequality. I test this theory by treating industries as miniature labor markets within the US with varying levels of gender inequality and different hiring practices. Using high quality administrative data from 1985 to 2013, including detailed work histories from this period, I compare the earnings of alike men and women across industries with different levels of reliance on external markets at different times.

I find that men experience greater unexplained earnings relative to women – unexplained in that it is not accounted for by work history or observable demographic characteristics – when a greater share of earnings increase events occur outside the firm.

Prior research has focused on within-firm dynamics relationship to gender inequality. Here, I shift the focus to between-firm dynamics. This shift is necessary, because it mirrors the shift in the labor market. I do not, in this chapter, offer a *causal* explanation of why external hiring produces a larger gender wage gap in an industry – though, I discuss existing theories that suggest explanations. Instead, I document an association at the industry level between prevalent hiring or job-filling practices and gender difference in earnings.

## Hiring Strategies and the Gender Earnings Gap: Explanation from Existing Theory

The gender earnings gap remains large and is no longer declining: though women's earnings were rapidly catching up with men's up to the 1980s, progress has slowed, even stalled, and where it has happened it has been uneven (Blau and Kahn, 2008, 2017; Gharehgozli and Atal, 2020).

There is a decades-long literature decomposing the sources of this gap. I briefly review the key contributions of this voluminous research.

First, women's labor force participation remains lower than men's. Women rapidly entered the US postwar workforce (Goldin, 2006), but workforce attachment was uneven, with wives of higher-earning spouses most likely to enter (Autor et al., 2008). The differential labor force participation and attachment of men and women introduces a selection problem. It could, for example, be that women who enter the labor force have higher expected returns to skill (as findings by Autor et al., 2008, suggest), and hence we may expect to see *higher* wages for these women than for men. Though researchers have used different strategies to account for this problem, none is perfect, and the selection effect ought to be borne in mind (see, e.g., Blau and Beller, 1988; Mulligan and Rubinstein, 2008).

Second, education and skill offer an intuitive reason for earnings differences. Yet, differences in educational attainment cannot explain the gender wage gap, as women in the US – and much of the developed world – earn more than half of degrees awarded (Buchmann and DiPrete, 2006; Buchmann et al., 2008). Some difference in field of study remains, with women under-represented in potentially lucrative Science, Technology, Engineering, and Math fields (Ceci et al., 2014; Cimpian et al., 2020; Lundberg and Stearns, 2019). Recently, attention has been given to gender difference in soft skills, learnt social-cognitive abilities (Heckman and Kautz, 2012), and psychological attributes (Buser et al., 2014; Nguyen et al., 2022; Niederle and Vesterlund, 2007). These avenues of research are interesting, but it is not clear that women's relative risk-aversion, for example, would be a disadvantage in all occupations, or that men's competitiveness should always produce an advantage.

Third, the gender wage gap could be explained if women have less experience and work fewer hours than men do. Supporting this explanation, Bertrand et al. (2010) find that the gender pay gap for MBA graduates is mostly explained by the women having less experience and hours post graduation. These women do not have long employment gaps, but even short lapses in employment carry a large penalty (though note Pedulla, 2016, finds a larger penalty for nontraditional work

histories for men). Goldin (2014) identifies this problem as one of *temporal inflexibility*, where workers face relatively severe wage penalties for not being at an employers' constant disposal. For women, the inability to be as available as demanded is often tied to increased responsibilities outside of paid formal work.

Family status presents another explanation for the gender gap in earnings. There is a well documented effect of motherhood lowering wages as well as lifetime earnings (Correll et al., 2007; Gough and Noonan, 2013). Further, in dual-earner households, women are likely to be the secondary earner, and family migration decisions are likely to be made based on a man's career, disadvantaging womens' earnings (Shauman, 2010; Shauman and Xie, 1996). Bertrand et al. (2015) find that marriage rates go down when women outearn men, suggesting that gender norms in household earnings distribution could also be suppressing women's earnings.

Closely intertwined with these explanations is the effect of employer demand-side discrimination. Numerous experiments have found discrimination in the form of lower offered starting salaries, better grades, smaller raises, and fewer promotions for women (Correll et al., 2007; Goldin and Rouse, 2000; Moss-Racusin et al., 2012; Neumark et al., 1996; Quadlin, 2018). These findings are not universal, with women job candidates preferred in certain settings (Williams and Ceci, 2015), but they document a process likely to explain some of the unbridged gender gap.

Finally, and accounting for a lionshare of gender earnings inequality (Blau and Kahn, 2017), women work in occupations and industries which pay lower wages (Blau et al., 2013). Levanon et al. (2009) explore two explanations for why occupations with a greater proportion of women see lower wages. The first hypothesis is that employers have a preference for men employees, and thus a gendered labor queue is formed. Women are paid less, because they are not men, and their great share in the workforce of that occupation results in lower average wages. Supporting this explanation is the presence of "glass escalators" for men in woman-dominated occupations (Williams, 1992). This explanation, then, becomes one of discrimination. The second hypothesis is that women's work is devalued, and thus pay is low in these occupations. Levanon et al. (2009) find strong evidence for the devaluation explanation.

**Firms as Wage Setters**

The role of the *firm* as a wage setter is missing from these dominant accounts of the gender earnings gap. Yet, firms are important for wage determination. A broad literature documents that some firms pay systematically more than others for the same tasks performed (Card et al., 2018, 2013; Song et al., 2019). An increasing number of scholars are focusing on firms' roles in the gender earnings gap (e.g. Blau and Kahn, 2017; Bruns, 2019; Card et al., 2016; Cruz and Rau, 2022; Li et al., 2022). This focus on firms adds a new perspective to understanding gender earnings stagnation. First, I describe how a firm-wage premium occurs. Second, I discuss why the magnitude of this firm-specific premium might vary across industries. I then turn to theorize how this affects the gender earnings gap.

There seem to be constraints on inequality within firms that prevent high-productivity high-wage firms from paying lower productivity workers or workers in lower productivity tasks much less than others – which may explain why firms are pushing for outsourcing of such lower productivity tasks (Goldschmidt and Schmieder, 2017). In fact, wage inequality within firms is sufficiently constrained that when a wage change occurs in the US, it has significant consequences on the renumeration of similar tasks performed in the same multinational companies *outside* of the US (Hjort et al., 2019).

But if wages are constrained to display relatively little variation between the top and the bottom performers within a company, paying your top performers more could have a significant further cost: wages for lower performers and lower productivity tasks go up. Thus, wages will show an element of *compression*; top performers make less, low performers make more (Leonardi et al., 2019). A direct consequence of compression is that firms which do not employ many workers in lower productivity tasks can compete favorably for top talent as they can raise wages for top performers at a lower total cost. Thus, as Gilley and Rasheed (2000) put it, firms that outsource can "make more by doing less". The theoretical question then becomes: why do firms ever directly employ workers in the lower productivity tasks? One reason may be that talent is hard to

observe for young workers and thus it may pay off to employ many such workers to learn who will become a top performer. Firms who do not engage in hiring at the bottom of the (observed) skill distribution will have to hire these workers from outside, potentially at higher costs. Second, there may be valuable firm specific skills that can only be learned by working your way up the skill chain – in traditional manufacturing companies, many of the top management and engineers used to be "born on the factory floor."

Thus, there are two features that would push firms to have a wider skill and pay range: 1) the ability to identify talent (which may differ between firms and between industries), 2) the necessity to train talent from the bottom up. Investments in firm-specific capital limit employee mobility (Campbell et al., 2012; Wang and Barney, 2006). The necessity of making these investments varies substantially by industry. On the other hand, the ease at which firms can poach top performers from each other will reduce the willingness of firms to engage in the full wage spectrum. Industries in which, for cultural reasons or because of the importance of firm specific skill sets, poaching is hard to accomplish, firms will face fewer costs of underpaying their top performers. In other industries, in which top performers can easily change firms, paying them less in order to maintain wage compression will be much more costly in terms of talent lost.

In a highly stylized sense, these competing factors lead to different topographies of firms in different industries. In some industries, firms will tend to span the full wage and skill spectrum, thus competing with each other for talent parallel along the entire spectrum. In other industries, firms will tend to span a sliver of the skill spectrum and layer themselves vertically, with firms competing for the top performers from the firms below them in the skill stack.

**Poaching Rather Than Promoting**

A corollary of wage compression is that firms increasingly hire employees from competitors to fill positions, rather than training up their own employees (Bidwell et al., 2013). The extensive use of outsourcing and firm specialization means the fabled trajectory from mail-room attendant to CEO would probably require a number of employer changes (even if you never really left the building).

Employers also value skills, experience, training, and prestige acquired in different employment contexts (Bidwell et al., 2015). There may also be more nefarious reasons for the appreciation of external talent. You need not invest so much in the health and well-being of an employee you only plan on having around for a few years. You do not want to burn out the "company man" in who you have invested so heavily, but it matters less to the company if an employee fizzles out just as they are about to leave for their "next great challenge" (Bidwell and Mollick, 2015; DeVaro et al., 2019).

There are also consumer-reputation perks to not training up your own employees. For example, budget airlines receive criticism for overworking and underpaying their flight attendants. Other carriers underscore the superior service provided by their high-quality well-paid crew members. What this misses, of course, is that the budget airlines provide training and initial screening of flight attendants – their experienced employees then get offered more lucrative positions at higher-end airlines.

External hiring can also been seen as a *worker-driven* product of low employee retention as more employees feel comfortable exiting employment. This narrative has become particularly forceful in the wake of the Great Resignation of 2021 (Gittleman, 2022; Kochhar et al., 2022; Parker and Menasce Horowitz, 2022). Firms are compelled to hire externally – often at higher wages (Bidwell and Mollick, 2015; Kochhar et al., 2022) – to account for their low employee retention. Though there is truth to this, it is nevertheless important to not overplay the ability of most workers to leave an employer they are not happy with. Further, the supply and demand sides of jobs and workers are naturally intertwined. For a worker to exit a firm, there must be another firm happy to hire externally.

There is significant debate about why firms would choose (if they choose) to poach externally rather than promote internally and what the effects are on employee performance (Bertheau, 2021; Bidwell, 2011; Keller, 2018). Nevertheless, employee churn has certainly increased overall in the U.S. economy (Andersson et al., 2005; Bidwell and Mollick, 2015; DeVaro et al., 2019; Wilmers and Kimball, 2022). This means that more employees face the hiring stage of the job search

process more frequently.

### Women in External Labor Markets

These changes in career trajectories have implications for women in the labor market. There is a difference in the expected amount of discrimination women experience between the hiring and promotion stages. Petersen and Saporta (2004) provide a framework of the opportunity structure for gender-based discrimination in an organization. The authors analyze five processes or career junctions that impact the allocation of workers: initial job level and wage, subsequent job level and wage, promotions, glass ceiling, and departure from the firm. Their framework suggests that most discrimination would happen at the initial job level. This is where the costs to the firm are lowest. While discrimination at the point of hire is also illegal, it is not as visible as when a man and woman already employed by the same organization are treated differently. Women not hired by the firm are also less likely to pursue legal action than those who already work in the offending organization. In contrast, the authors find little evidence of a "glass ceiling effect", where women already in the organization get passed up for promotion. Kalev and Dobbin (2006) offer corroborating evidence: they find that the enforcement of civil rights law does little to improve the employment rates of women and African-Americans. Though employers formally comply with the law, the intended consequences – equal employment for disadvantaged groups – do not materialize.

To complicate matters, a set of rules created to protect a disadvantaged group in one of these evaluative situations might have spillover effects into the other. For example, imagine that there is a firm in which biased hiring creates a double standard for men and women: women are held to a higher performance standard before they are hired. In this situation, newly hired women have on average higher performance than newly hired men. We might then observe that women have higher promotion rates than men, because the women at this firm are simply more skilled on average. Now, imagine this firm introduces new protective measures in its hiring practices, and that doing so removes the double standard. We would expect to see more women among new

hires, but, at the same time, the average skill of the women hired would be lower (i.e. equal to men's) than under biased hiring. In this situation, we would observe a reduced promotion rate for women within the firm. A change in hiring practices could in this way have an effect on promotion outcomes, despite no change in promotion practices.

Internal labor markets too leave opportunity for unequal outcomes and biased evaluation. Castilla (2008) documents a *performance-reward bias*: minority employees within a firm are not equally rewarded for the same performance (as evaluated by their superiors) as white men are. Pekkarinen and Vartiainen (2006) find that men are favored for promotion over women, with the average promoted woman demonstrating much higher pre-promotion productivity than the average promoted man. More subtly, Petersen and Saporta (2004) describe a theory of allocative discrimination, where the main impediment to women's internal labor market mobility is their allocation to low-skill or low-prestige tasks. Similarly, research by Baldi and McBrier (1997) suggests internal labor markets disadvantage Black employees inside the firm. The authors surmise that this effect might be because Black employees are allocated tasks that are low-skill or low-prestige, compared to white employees in the same vertical position. When employees are not allocated interesting or challenging tasks, they do not have the opportunity to earn promotions (Jones and Makepeace, 1996). Thus, even when employees might be fairly *evaluated* within firms for internal promotions, they may by unequal in their ability to exhibit their value to the firm by working on challenging tasks.

Barnett et al. (2000) argue these "work roles" go further in explaining job mobility. Women and nonwhite employees work in positions that require general rather than firm-specific capital. Perhaps this is because they are not allocated interesting tasks important to the firm. There should then be higher rates of external labor market activity for these groups (because they have no firm-specific skills they are giving up). Among managers, Brett and Stroh (1997) find that men receive higher compensation when they move between firms, but women do not. Similarly, Quintana-Garcia and Elvira (2017) find that women executives who are hired in an external labor market are disadvantaged in earnings. Dreher and Cox Jr (2000) find that gender and race moderate

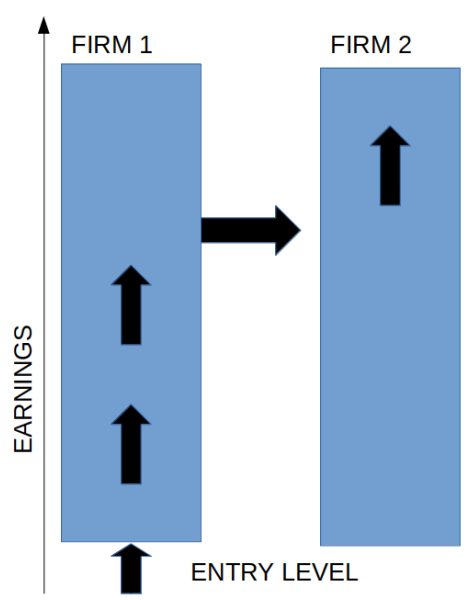
the earnings effect of switching between employers. Fuller (2008) suggests frequent employer changes have only mixed effects for women's earnings. Kronberg (2013), on the other hand, finds that gender disparities in earnings decrease among voluntary leavers of good jobs. These mixed findings suggest that the reason, or perceived reason, behind the employer changes could affect earnings. Frequent job changes can be taken to signal a dilettante or a renaissance man (Leung, 2014) – the interpretation of this type of job history could well be gendered.

These studies do not observe the reason for the job change. For example, if women are more likely to change employers for non-economic reasons (e.g., they are relocating for a spouse's job, see discussion in Chapter 3), we would not expect to see a compensation advantage. We might even expect this is a reason women invest in general skills, if they want to be prepared to switch employers. It is also reasonable to expect that women might place greater value in non-pecuniary forms of compensation, such as freedom from harassment (Hart, 2021). This too might cause us to observe job changes that do not lead to earnings increases.

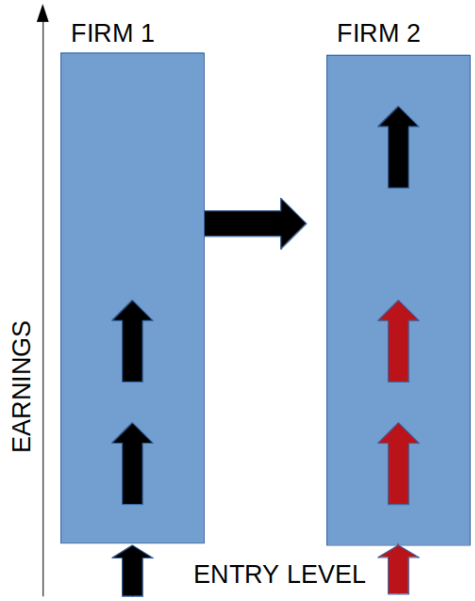
### Hypotheses

The arising hypotheses are best expressed visually. Figure 1a presents an industry with two firms. Each arrow represents an earnings increase event. A man enters firm one (represented by the small black arrow) and experiences two promotions (represented by the thick black arrows). After this, he experiences a lateral move to firm two, where he again gets a raise. This small industry has very little movement between firms – and, in fact, only one lateral move. Figure 1b shows a woman entering this same industry. She experiences the same two promotions as the man, but hits the proverbial glass ceiling when a man is hired above her.

Figure 2a presents an industry of two firms with more between-firm movement. Again, a man enters firm one. He is immediately offered a more lucrative position in firm two (designated by the diagonal arrow). He then receives an offer to return to firm one at equal pay (this unrealized offer is represented by the grey arrow), but instead he leverages this for a promotion at firm two. Even in this highly external-market-driven industry, promotions do happen, but they happen in




(a) An industry with low reliance on external hiring, a man's trajectory in black




(b) An industry with low reliance on external hiring, a woman's trajectory in red

Figure 1: Example industry with **low** reliance on external hiring

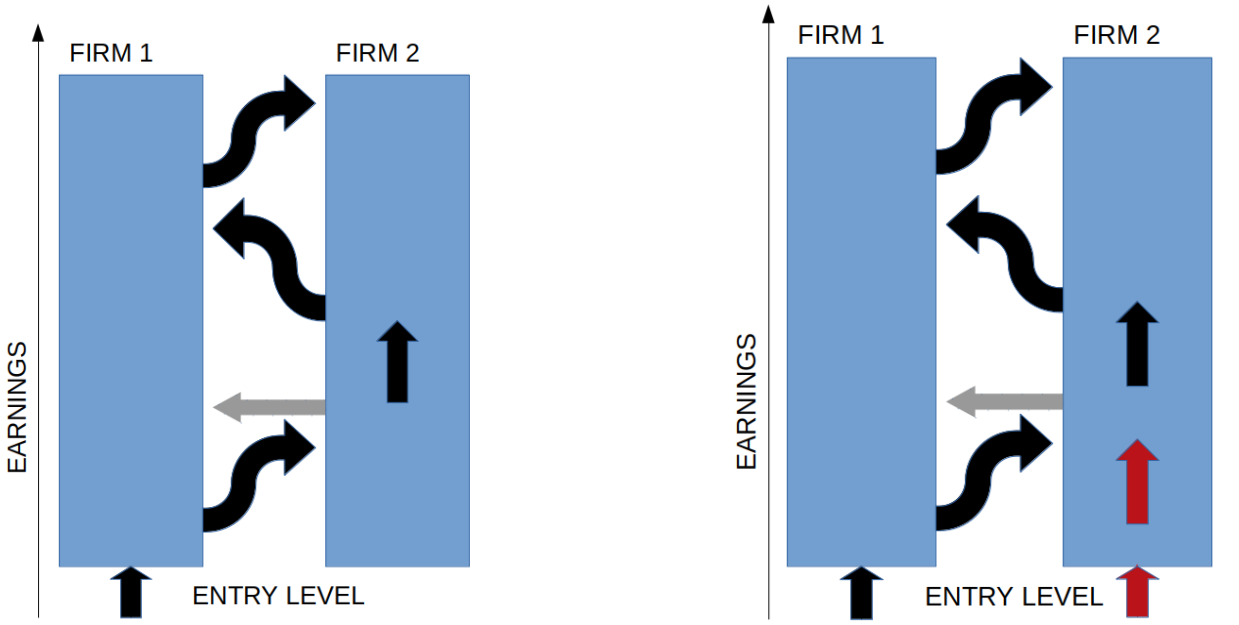
response to external market success. The man is then offered a more lucrative position in firm one, from where he again switches back to a higher paid position in firm two. Figure 2b shows the short stagnated career of a woman in this industry. She enter firm two, but does not gain entry to firm one, and thus cannot use the between-firm competition for talent to see her earnings increase.

Thus, the hypothesis that emerges is that the gender earnings gap will be greater in industries with greater reliance on external hiring. In industries where internal promotions are the more dominant pathway to earnings advancement, women bridge the earnings gap.

### Why Industries?

The patterns I have described are economy-wide. Yet, in this chapter, my unit of analysis is the industry. The focus on industry creates higher level groupings that allow me to study internal and external labor markets.

Within an industry, a large variety of careers are possible, as everything from the actual production work in the industry, managerial roles, marketing roles, human resources, legal services,




(a) An industry with high reliance on external hiring, a man's trajectory in black

(b) An industry with high reliance on external hiring, a woman's trajectory in red

Figure 2: Example industry with **high** reliance on external hiring

logistical support, and myriad other ancillary services are all included in the same industry as long as they occur at the same establishment. Nonetheless, there is a strong reason to consider an industry more than a meaningless statistical designation from the perspective of the individual employee – the method of production and the production process is apt to shape what companies are meaningful competitors not just for customers but also for employees. Employees are more likely to know their counterparts at a rival firm in the same industry than their counterpart at a completely unrelated firm. This and other vectors for dissemination of industry culture and norms mean an industry is a meaningful area to determine how career advancement occurs for an individual employee.

Analyzing the industry level also helps avoid many issues of employee self-selection into types of labor markets that would arise with focus on the individual level or on occupations.

### A Note on Terminology

In my above description of two earnings trajectories, I have remained intentionally vague about the impetus for each earnings increase or firm change. I do not observe whether a move was worker-driven, e.g. if a man applied to a new job to be closer to family, or employer-driven, e.g., a headhunter approached him with a lucrative offer. These are at first blush different processes for which we use different words, such as job-seeking compared to employee-poaching. Certainly, it feels different to be a worker trying to secure a new job than to be approached by recruiters regularly. There are also important push-pull factors: is a firm change happening because the new firm has nicer benefits, or because the old firm had a hostile work environment? However, though it is interesting to study the demand and supply sides of job search, it is also important to remember that they are deeply endogenous and jointly determined by broader labor market conditions. Thus, an attempt at separating out these two processes in real-world data would likely be artificial and contrived.

In this chapter, I will use terms related to job change fairly interchangeably. I sometimes refer to a firm change as a "poach", though I have no information on whether the new firm truly reached out and poached the employee in the usual sense of the word. I also use the word "promotion", but want to remain indifferent to whether or not there was a change in job title. More accurately, I am interested in *earnings increase events*. I observe large discrete jumps in earnings and whether those occur within a firm or across two different ones. However, though I remain agnostic in any given case for why an earnings increase event occurred, I use more specific terms to remind the reader of the types of processes underlying these observations.

## Data and Methods

The data for this analysis come from the Longitudinal Employer-Household Dynamics panel (LEHD). The LEHD is a US Census Bureau created data set that combines information in W-2 form with other US Census information such as age, race, and gender. The LEHD is at individual-

quarter level and individuals are automatically linked across quarters and across sources of W-2 earnings within a given quarter. To compute an individual's total earnings over a given time-span, we could sum across W-2 earnings for all of her different employers in the given quarter-range.

The sample spans a subset of US states with some states entering the sample at a later date than others. For quarters in which a US state is included in the sample, one can observe the *entire* uncensored earnings information for every single individual who worked in the state in the given quarter. Due to the sheer size of the data (quarterly observations on a vast majority of wage earners in the US for multiple decades), I use a random 10 percent sample from 1985 to 2013 – the size of the sample has no qualitative effect on results due to the reliability with which population coefficients can be estimated at this magnitude of data points.

I report all analyses at the *industry*-quarter level, i.e., the raw individual level data is aggregated to industry-level information and regression models are estimated based on industry-quarter observations. The definition of industry in the data is described below. The choice to aggregate to the industry level follows from the theoretical arguments raised in the previous section: the hypothesis is that women and men differ in their earnings-trajectories *within* an industry based on how the given industry's career advancement path functions. Analyses at the individual level thus do not yield any additional insight as all observations within a given industry face the same explanatory variables. There are 102,000 industry-quarter pairings. These are based on 4,240,000 unique firms and 12,110,000 unique people.

Administrative W-2 based tax data have many advantages. These data provide the most accurate and comprehensive information on earnings, without being plagued by concern over recall, misunderstanding of what constitutes earnings, mis-statement to hide or exaggerate earnings to the enumerator, censoring or top-coding (which is common in publicly available survey information) and self selection (Angel et al., 2019; Bound and Krueger, 1991). However, there are certain limitations to what even this type of data can tell us. Specifically, non-traditional employment is not captured accurately, self-employment is absent (albeit also not theoretically salient to the question at hand), and of course any off-the-books earnings are invisible, too. Seasonal work and multiple

job holdings are included in the data but it is not recorded whether a given job is seasonal (and a job ending was pre-determined or whether it was an actual quit or layoff) nor do we know whether earnings from two or more employers in a given quarter occur concurrently or sequentially.

Moreover, demographic information is somewhat limited by what other US Census Bureau information can be linked to W-2 information. Gender and age are reported very accurately and with very high degree of coverage. Race is also reported for the vast majority of individuals but is not as near universal as gender. I exclude people whose gender is missing or imputed and code missing race information as its own race-category for analyses that include race. In theory, one could try to condition models on educational attainment, as some individuals have information on their education recorded in the LEHD, but coverage is poor and data quality is uncertain. I thus elect to constrain conditioning variables to the high-coverage, high-quality race and age covariates.

Another limitation of the earnings information is the absence of hours information on W-2s. Given a specific earnings data point, one cannot tell whether an individual works full-time at the given employer, part-time at this employer, or full-time but only for some weeks of the quarter at this employer. The theoretical model posited above only truly applies to full-time employees, as the earnings dynamics of part-time employees are significantly more complex – a large earnings fluctuation can arise from changes in hours rather than changes in job title or function. In order to remove part-time workers in the data, I exclude employer-quarter observations that fall below a given earnings threshold that proxies for part-time status.[1] While I exclude part-time quarters from the analytical sample, note that I do include part-time quarters in constructing my measure of work experience at the individual level. The lack of hours data is not of great concern to my analyses. Theoretically, it is possible that there is significant gendered variation in hours worked across industries. It is unlikely that this is correlated with the ratio of promotions. Specifically, given that I conduct analyses looking at changes in ratios of promotions *within* industries, it is exceedingly unlikely that my findings could be driven by a spurious correlation with work hours.

All regression analyses in this chapter seek to illuminate the question of whether variation

in the unexplained earnings disadvantage of women can be explained by variation in industries' reliance on firm-external labor markets over firm-internal ones. The regressions thus all place a metric of women's unexplained earnings disadvantage – specifically the residual gender earnings gap – on the left-hand-side while using a variety of measures of the use of external labor markets on the right-hand-side, in addition to control variables. I introduce all of these measures and variables' construction and theoretical properties in the subsections below.

**Unit of Analysis: Industry**

The statistical construct of an industry requires classifying "similar" firms as belonging to the same industry grouping. What counts as similar depends primarily on the needs of the data-user that had been kept in mind when designing the original industry-classification system. In this paper, I use the North American Industry Classification System (NAICS) throughout. The NAICS has been the standard used by Federal statistical agencies in classifying business establishments for the purposes of statistical data related to the U.S economy since 1997 and was developed under the auspices of the Office of Management and Budget (OMB).[2]

The NAICS classification system groups establishments by their process of production. Firms that use similar production processes are grouped in the same five digit industry coding. The NAICS code thus would assign 31121 to a corn flour milling establishment; composed of first the two-digit code for food manufacturing (31), then the industry-group code for oil and seed milling (12), and finally the last digit (1) for the specific sub-industry of flour milling. A garden-design firm would be assigned the code 54 for the professional services sector, 13 for design services, and industry code 2 for landscape design services, an industry grouping it shares with ski-resort planning services, landscape architects, and golf course architectural bureaus.

**Dependent Variable: Residual Gender Earnings Gap**

I seek to capture the theoretical construct of a difference in the ability of men and women to advance their careers in different industries based on how career-progression channels operate in

a given industry, namely, whether between-firm or within-firm advancement is more prominent. To approximate the difference in career advancement, I compare the earnings of men and women who work in the same industry, in the same quarter, with identical work experience, and identical other demographic information (age, race). Any earnings difference that remains can be attributed to the given man and woman to have enjoyed a different career progression up to this point. While any two men and women may have vastly different earnings absent any discrimination or structural differences in work opportunities, the *mean* difference between men and women of identical observable characteristics in a given industry-quarter may be indicative of a structural difference in the opportunities for career advancement. I thus need to compute the *mean residual earnings gap* between men and women in the given industry-quarter.

The dependent variable for all of my analyses is thus the mean residual gender earnings gap in an industry in a given quarter. Calculating the residual gender earnings gap at the industry-quarter level takes three steps. First, I predict earnings at the individual level using a linear regression of earnings on a polynomial in age, a polynomials in work experience, industry-experience, and length of employment spell, race indicator variables ($r \in \mathcal{R}$), and state fixed effects ($s \in \mathcal{S}$). This regression at the individual level does not include gender as an explanatory variable, as I will later calculate statistics on the aggregate level by gender. In short-hand, I run a model of the form:[3]

$$\text{Earnings}_{i,t} = \text{age}_{i,t} + \text{work history}_{i,t} + \text{race}_i + \text{state}_{i,t} + \epsilon_{i,t}$$

Second, I compute the difference between the actually observed earnings for a given individual $i$ in a given quarter $t$ and their predicted earnings, i.e., I find the residual earnings $\epsilon_{i,t}$. A positive residual indicates that the individual earned more in the given quarter than the average individual of identical age, race, work experience, in the given state in the given quarter.

$$\hat{\epsilon}_{i,t} = Earnings_{i,t} - \widehat{Earnings}_{i,t}$$

Third, I compute the mean of the residual earnings at the industry-quarter-gender level, i.e.,

for every industry $m$ I find the mean residual for men ($i \in \mathcal{M}$) and women ($j \in \mathcal{W}$ )in this given quarter, and take the difference between women's mean residual and men's mean residual. A negative mean residual earnings gap thus indicates that men earned more than women in the given industry in the given quarter, relative to their observable characteristics such as work experience.

$$Y_{t,m} = \frac{1}{|W_{t,m}|} \sum_{i \in W_{t,m}} \hat{\epsilon}_i - \frac{1}{|M_{t,m}|} \sum_{j \in M_{t,m}} \hat{\epsilon}_j$$

To summarize: a non-zero residual earning implies a premium to earnings that cannot be explained by the observables, i.e.., variety of work experience measures, age, state, or race.[4] If men have on average larger unexplained premia to their earnings than women, for example due to the hypothesized opportunity to switch jobs for career advancement, then this would reflect into a negative mean residual earnings gap for a given industry and quarter.

**Explanatory Variable: Measures of External Earnings Increase Events and Employee Churn and Movement**

The theoretical model posits that an industry that relies more on internal promotions rather external hiring for career advancement offers greater protection to women against bias and discrimination. Thus, to adequately test the theory, I need to define variables that capture the prevalence of promotions compared to external hires as a means of career advancement. The goal is to use a variable that 1) captures the sociological concept of "how important" it is in an industry to apply to other companies to receive a pay increase, 2) but not spuriously create results by capturing other industry features such as, e.g., employee fluctuation or "churn" differences between sectors of the economy, secular growth or decline of an industry, or seasonal or non-standard work. I choose to concentrate on the *ratio of earnings increase events* that occur due to an internal boost (promotion) versus all earnings-increase (promotion and poach) events.

**Ratio of internal promotions to all earnings increase events**. I contrast how many earnings increase events happen within the same firm, compared to those that happen when a worker

changes firms. We may think of within-firm earnings increase events as classical raises or promotions, while earnings increase events that occur after a firm-switch are being hired for a more senior – or at least better remunerated – position than one's current position. Note that it is not relevant that the promotion actually come with a formal change in job title or specific symbols of seniority – what matter is the actual economic impact, in terms of earnings, on career advancement.

I use the ratio of internal (within-firm) to all (external + internal) earnings increase events in order to capture the reliance of an industry on firm switching for career advancement, without accidentally capturing pure scale or growth effects. The latter might contaminate my results if I used a level-based or absolute variable rather than a ratio-based variable. For contrast, consider the alternative case of computing the number of external earnings increase events per employee. In some industries there are more earnings increase events per employee than in others, external or internal. If women and men differ by their propensity to work in fast-advancement industries, I might capture this difference when I mean to capture a reliance on external hires if I used the number-of-events variable instead of my preferred ratio-of-events specification.

One key challenge is to define earnings increase, a discrete event, in terms of the available continuous earnings variables. Prior research has typically determined a promotion to have occurred when a worker sees a discrete increase in earnings from the previous quarter, and this increase persists into the next quarter. This is important, because it removes variation that could come in from e.g. seasonal workers taking on more shifts or hours, or from one-off bonuses. I present results using two different thresholds for an earnings increase event to count as a "promotion" to ensure that my results are not driven by accidents of a given threshold. For my main analyses, I use a 10% threshold to define a discrete earnings increase event, which is common in the existing literature. This means that a worker is determined to have experienced an earnings increase event this quarter if her earnings are 10% higher this quarter than they were the previous quarter *and* these earnings do not decrease in the next quarter. The condition that earnings increases are persistent is important to exclude quarters in which employees received seasonal bonuses or otherwise

received temporary increases. To check the robustness of my findings and make sure they do not depend on this specific choice of threshold, I present supplemental findings using a 20% threshold in Appendix A.

An "internal earnings increase event" or internal promotion, then, is when a worker experiences an earnings increase event across quarters where she has worked for the same firm. A worker cannot experience an earnings increase event in the first or second quarter of working for a new firm. An "external earnings increase event" occurs when the worker stops working for one firm and begins work for a new firm (new in the sense that they had not worked there in the previous quarter – I do not exclude cases where a worker had at some point worked there before). I do not include the quarter where a worker changes firms, because I do not know the precise date at which the new job started, and thus cannot appropriately prorate earnings.

My key explanatory variable is thus the ratio of internal promotions to all earnings increase events at the industry-quarter level. The higher the number, the more earnings increases occur within-firms rather than between-firms. This indicates that the industry is more reliant on internal promotion than it is on external hiring.

**Ratio of internal promotions to all earnings increase events, for men and for women.** I further study the effects of three additional explanatory variables that highlight different features of the sociological theory I seek to test. First, I separately regress the residual gender earnings gap on the ratio of internal promotions to all earnings increase events for men *and* the analogous ratio for women. If industries not only differ in their reliance on different job filling strategies but also in how equal access to such events is for men and women, this specification would allow for both differences to predict increases in the residual earnings gaps separately.

**$\Delta$ Ratio Men-Women.** I also use a variable, "$\Delta$ Ratio Men-Women", to capture the difference of the gender-specific ratio variables.

**Proportion of new hires.** Finally, I will also consider an extremely simple measure of reliance on external hires: the proportion of new employees at a given firm in a given quarter, i.e., which of the current employees were not employed at the firm in the previous quarter. This simpler

measure does not capture the theoretical construct we seek to illuminate as neatly and may be contaminated by other, unrelated employment dynamics, such as a faster growing industry also having more new employees, as all companies are hiring new workers. Nonetheless, it offers an advantage over the previous statistics in that it encapsulates a very simple and broad metric of external hiring frequency. This most simple measure of how many workers face the hiring stage in an industry is the proportion of new employees in a given quarter. Employee hires are given by the number of employees who did not work at the given firm in the last quarter and now have reported earnings in this given quarter, over the average number of old and new workers at the given firm.[5]

Note that this definition does not match the standard measure of employee churn (the number of exactly offsetting hires and separations), nor that of job growth (hires above separations), see e.g. Lazear and Spletzer (2012) or Hyatt and Spletzer (2013). For my purposes, what is interesting is how many people are simply new at a given company. Table 1 shows summary statistics of the key variables in the data.

### Control Variables

Despite selecting an explanatory variable – the ratio of internal to all earnings increase events – that is designed to focus on the key sociological construct that we seek to illuminate, this variable may still be correlated with other extraneous aspects of an industry's earnings dynamics that could obscure the actual relationship that we seek to find. Thus, I construct a number of control variables to soak up unrelated variation in our outcome of earnings growth.

**Growth**. First, I compute the growth rate of an industry by computing the net of new hires and quits over the average of current and last quarter's total employee numbers.[6] Controlling for growth rates is important as it is plausible that men and women may differ in their propensity to work in fast-growing industries, fast-growing industries may differ in how earnings develop over a career, and fast-growing industries may have more external promotions (Sorkin, 2017). Similarly, declining industries may differ from steady industries, too.

| Variable | Mean | SD | 5th | 25th | 50th | 75th | 95th |
|---|---|---|---|---|---|---|---|
| Residual Earnings Gap | -5,882 | 5,236 | -14,800 | -73,31 | -4,810 | -2,919 | -377 |
| Prop. New Hires | 0.178 | 0.127 | 0.047 | 0.094 | 0.147 | 0.224 | 0.414 |
| Prop. Int. Promotions (10%) | 0.086 | 0.057 | 0.015 | 0.048 | 0.075 | 0.11 | 0.194 |
| Prop. External Increase Event (10%) | 0.078 | 0.078 | 0.017 | 0.041 | 0.061 | 0.085 | 0.194 |
| Ratio Int./Tot. Events (10%) | 0.453 | 0.215 | 0.129 | 0.305 | 0.434 | 0.579 | 0.867 |
| Ratio Int./Tot. Events, Women (10%) | 0.453 | 0.239 | 0 | 0.296 | 0.438 | 0.593 | 0.921 |
| Ratio Int./Tot. Events, Men (10%) | 0.452 | 0.225 | 0.11 | 0.298 | 0.432 | 0.586 | 0.884 |
| $\Delta$ Ratio Men-Women (10%) | -0.001 | 0.172 | -0.25 | -0.062 | 0 | 0.059 | 0.244 |
| Prop. Int. Promotions (20%) | 0.045 | 0.032 | 0.005 | 0.025 | 0.039 | 0.057 | 0.101 |
| Prop. External Increase Event (20%) | 0.076 | 0.078 | 0.017 | 0.04 | 0.06 | 0.084 | 0.193 |
| Ratio Int./Tot. Events (20%) | 0.583 | 0.206 | 0.234 | 0.465 | 0.586 | 0.712 | 0.936 |
| Ratio Int./Tot. Events, Women (20%) | 0.577 | 0.241 | 0 | 0.449 | 0.588 | 0.727 | 1 |
| Ratio Int./Tot. Events, Men (20%) | 0.584 | 0.219 | 0.2 | 0.457 | 0.589 | 0.723 | 0.962 |
| $\Delta$ Ratio Men-Women (20%) | 0.007 | 0.2 | -0.282 | -0.067 | 0 | 0.074 | 0.303 |
| $N$ unique workers: 12,110,000; $N$ unique firms: 4,240,000 | | | | | | | |

Table 1: Summary Statistics of Key Variables for Analytic Sample, industry-quarter-year level

Note: All variables are at the industry-quarter-year level. Percentiles are fuzzy and based on the 10 industry-quarters around the true percentile. The residual gender earnings gap is calculated by predicting earnings for each individual using work experience (how many quarters worked), part time experience (how many work quarters were part time), industry experience (how many work quarters were in this industry), age, age squared, industry, length of current employment spell (number of quarters worked consecutively leading up to this one), race, and state. "Prop. New Hires" is the proportion of new employees this quarter. "Prop. Int. Promotion" is the proportion of workers who experienced a firm-internal earnings increase event this quarter, using the 10% or 20% earnings increase definition respectively. "Prop. External Increase Event" is the proportion of workers who moved to another firm with a raise this quarter. "Ratio Int./Tot. Events" is the ratio of internal promotions to total earnings increase events, for the whole industry-quarter, or for men and women separately. "$\Delta$ Ratio Men-Women" is the difference of the gender-specific ratio variables.

**Gender composition**. I control for the percentage of men in an industry. By controlling for gender composition we are closer to the ideal-comparison of different promotion structures in otherwise "identical" industries. One worry is that industries with more women in them also have generally slower earnings growth. If industries with a greater fraction of women use external hires less frequently, e.g., because of other gendered components of the labor market such as women's lower propensity to move for a new job (see Chapter 3), we may then spuriously conclude an effect of external promotions on earnings advancement. Controlling for gender composition remedies some of this problem. Additionally, I control for the square of gender composition. Other researchers have hypothesized that the relationship between gender equality in pay and representation is curvilinear: e.g. if men experience a "glass escalator" in women-dominated industries while women are disadvantaged in men-dominated spaces (Budig, 2002; Williams, 1992).

**Advancement speed**. I construct a measure of the overall earnings advancement speed of an industry by computing the number of earnings increase events in the industry divided by the total employment number in the industry. I include this control variable for similar reasons as the gender composition control variable: I do not wish to spuriously conclude that a greater reliance on external earnings increase events causes a greater gender earnings gap if in truth men and women earn different amounts because women simply happen to work in industries with overall slower earnings advancement speed. This would also be an interesting sociological finding but not the mechanism I seek to illuminate in this paper.

**Industry fixed effects**. Industry fixed effects capture the time-invariant features of an industry that could contribute to the gender earnings gap. It is crucial to control for the fact that different industries have different levels of earnings equality – I would not want to conflate women working in industries that suffer from greater gendered earnings inequality in general with the specific mechanism I propose. When controlling for industry, the independent variables now reflect change in internal-external labor market hiring practice. In other words, they test earnings gaps associated with increases in use of external hiring, rather than the absolute level of external hiring in an industry. I run models both with and without industry fixed effects and my main findings are

robust to both specifications.

**Quarter fixed effects**. Quarter fixed effects net out any time-specific shocks that are associated with all industries in a given quarter. This allows us to compare industries across time without issues such as the overall narrowing of the gender pay gap complicating the interpretation of the results.

Note that throughout the paper, all values are shown to four significant digits.

# Results

I hypothesized that the gender earnings gap will be greater in industries with greater reliance on external hiring. And, conversely, in industries where internal promotions are the more dominant pathway to earnings advancement, women bridge the earnings gap. In the analyses that follow, the outcome variable is the residual gender earnings gap in the industry in a given quarter-year. Recall that the gender earnings gap is defined as the average residual earnings for women less the average residual earnings for men, after controlling for a host of career and demographic variables. A *positive* residual gender earnings gap thus means *higher* earnings for women relative to men, a *negative* gap means higher earnings for men relative to women – with the latter being the case for virtually all industries. The hypothesis of this paper thus can be stated as expecting an *increase* in the outcome variable, meaning a *narrowing* of the gender earnings gap, being associated with greater reliance on *promotions* rather than *poaching*. Thus, to empirically test this hypothesis, we consider multiple different explanatory variables that capture dimensions of this reliance on promotions versus poaching.

The explanatory variables are different ways to capture an industry's reliance on internal promotions versus external poaching for career advancement: 1) the ratio of internal promotions to all earnings increase events, 2) this ratio separately computed for men and for women, 3) and the difference between the promotion ratios for men and women. Further, I also regress the earnings gap on simply the proportion of new hires in an industry in a given quarter to check whether this even

more basic feature of an industry captures any relevant variation in the gender earnings gap. Thus, in the models including measures 1) through 3) above, my hypothesis would predict positive coefficients: the greater the share of earnings increase events internal to a firm, the smaller women's earnings disadvantage; the more *men* rely on internal promotions, the better for women's earnings relative to men; and the more men rely on internal promotions relative to women, the smaller the gap between men and women in earnings.

I examine this set of associations for three different model specifications, progressively controlling for more industry features that may attenuate or explain observed earnings differences. In Table 2 I show the results of *unadjusted* models, which include only these explanatory variables as well as a constant term. In Table 3, I introduce the industry-level control variables discussed above: employment growth, advancement speed, and gender composition, as well as quarter-year fixed effects. The control variables pick up on many previously hypothesized or reasonable other mechanisms that could drive gender earnings gaps and could be correlated with my main measures of internal promotion reliance. The quarter-year fixed effects eliminate time-trends in the residual gender earnings gaps. In Table 4, I include quarter-year fixed effects and industry fixed effects alongside the key explanatory variables. Controlling for industry fixed effects eliminates any time invariant features that could be driving gender earnings differences between industries. I discuss the results of each set of analyses in turn.

**Residual Gender Earnings Gap and Firm-Internal vs. Firm-External Earnings Increase Events**

I start the analysis with the uncontrolled associations between the career advancement variables and the residual gender earnings gap. Table 2 shows the first set of analytic results. The outcome variable in these models is the residualized gender earnings gap. Thus, a positive coefficient here means that the residualized gender earnings gap increases in favor of women. Given the *negative* baseline value of this variable in virtually all industries, this means the gender earnings gap narrows. All models are at the industry-quarter-year level. In Model 1, the explanatory variable is the

proportion of new workers, with no further controls. This model shows no statistically significant association between the proportion of new hires in an industry-quarter and the residual gender earnings gap. Hence, we have no evidence for simply the proportion of new hires already driving gender earnings differences.

In Model 2, the explanatory variable is the ratio of internal promotions to all earnings increase events, with no further controls. This model shows that as the ratio of internal earnings increase events increases – i.e., an industry relies less on external than internal hiring – the unexplained earnings gap between men and women begins to close. The coefficient is $700.9$ ($p < 0.001$), meaning a \$700 per year decrease in the magnitude of the gender earnings gap per full unit of the dependent variable. To put the magnitude of $\$700$ per year into perspective, consider that the standard deviation of the explanatory variable is approximately $0.2$. This means we expect a woman working in an industry with a one standard deviation higher ratio of internal to all earnings increase events would be expected to make approximately \$140 more per year than a similar woman in the average industry. An alternative way to make sense of this coefficient is to consider that, for two identical women, one working in an industry at the $5th$ percentile of internal promotion ratios would expect to make $\$517$ less per year than the woman working in an industry at the $95th$ percentile of internal promotion ratios.

Model 3 shows this association broken down for men and for women: the outcome variable remains residual gender earnings gap at the industry-level, and the explanatory variables are the ratio of internal promotions to all earnings increase events *experienced by women* and the same ratio as *experienced by men*. Holding men's earnings increase event ratio constant, a one standard deviation increase in the proportion of *women's* firm-internal earnings increase events is associated with a *widening* of the gender earnings gap by approximately $\$239$. On the other hand, holding women's earnings increase event ratio constant, a one standard deviation increase in the proportion of *men's* firm-internal earnings increase events is associated with a *closing* of the gender earnings gap by approximately \$407. This finding is of substantial importance. As more of the earnings increases experienced by *men* occur externally, the greater the earnings gap in their

favor. However, as more of the earnings increases experienced by *women* occur externally, the earnings gap decreases, i.e., women's earnings catch up. This finding suggests that when women are able to navigate between firms, they experience earnings increases more commensurate with those of men, allowing them to bridge the gap. Another way to think about it is to consider that when external labor market strategies are not prevalent in the industry, men are not able to pull ahead by quite as much.

Model 4 shows the association between the gender earnings gap and the difference between the rates of internal promotions (over all other earnings increases) for men and women. There are no further controls. This model reinforces the results of Model 3: it shows that in industries where more of *men's* earnings increase events happen within the firm, the residual gender earnings gap is smaller. Together, Model 3 and Model 4 suggest that men's residual earnings advantage is greatest when men move between firms more.

**Robustness to Industry-level Controls**

Table 3 repeats the models of Table 2 and introduces the following control variables: growth, advancement speed, gender composition, and quarter-year fixed effects. In Model 1, the proportion of new hires is associated with a bridging of the earnings gap: a one standard deviation increase in the proportion of new workers in an industry is associated with a $\$774$ closing of the gender wage gap ($p < 0.001$). This suggests that in a higher churn industry, holding all other controls constant, the residual gender gap is smaller. This goes against the hypothesis that the gender earnings gap would be greater in industries where the workforce must face the hiring interface more regularly. In this model, industry growth, on the other hand, is associated with a increasing gender earnings gap. There is no association between advancement speed and residual earnings.

The coefficient for gender composition shows that a higher proportion of men in an industry is associated with a greater earnings gap. The positive square term, however, suggests this relationship is U-shaped. In other words, in industries dominated by men, the gender earnings gap is greater. Similarly, at the other end of the spectrum, in women-dominant industries, the few

| Residual earnings gap (women-men) | (1) | (2) | (3) | (4) |
|---|---|---|---|---|
| Prop. New | 427 | | | |
| | (514.7) | | | |
| Ratio Int./Tot. Events | | 700.9*** | | |
| | | (184.6) | | |
| Ratio Int./Tot. Events, Men | | | 1807*** | |
| | | | (257) | |
| Ratio Int./Tot. Events, Women | | | -998.1*** | |
| | | | (202.6) | |
| $\Delta$ Ratio Men-Women | | | | 1314*** |
| | | | | (206.5) |
| Const. | -5,958 | -6,199 | -6,247 | -5,881 |
| | (159.3) | (156.5) | (162.8) | (127.9) |
| $N$ | 102,000 | 102,000 | 102,000 | 102,000 |
| $R^2$ | 0.0001067 | 0.0008302 | 0.002958 | 0.001857 |

+ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 2: Association of residual gender earnings gap with proportion of internal earnings increase events

Note: Standard errors are clustered on industry. These are the 'unadjusted' models – there are no further controls not reported here. The dependent variable is the residual gender earnings gap. Model 1 shows the coefficient for the proportion of new workers (churn), Model 2 for the proportion of internal promotions of all earnings increase events. Model 3 shows this ratio separately for men and women. Model 4 shows the difference between these two ratios. In all models, an earnings increase is defined as an increase of at least 10% this quarter compared to last quarter, which is sustained (lasts for at least two quarters at the same employer).

men will make more money, supporting the concept of a "glass escalator" for men in women's professions. In industries with a more even gender composition, the residual earnings earnings gap is smaller.

In Model 2, with all controls included, we see that women bridge the earnings gap in industries with more internal promotions. A woman in an industry with a one standard deviation greater ratio of internal promotions to all earnings increases events expects to see $\$530$ more per year.

In Model 3, the explanatory variables are the ratios of internal promotions to all earnings increase events, separately for men and for women, with all controls. In this model, the coefficient for the ratio of men's earnings increases is $3,040$, meaning that a one standard deviation greater ratio of men's earnings occurring within firms is associated with a $654 per year boost to *women's* earnings ($p < 0.001$). On the flip-side, this suggests that in industries where men experience more of their earnings increase events *outside* the firm, men's earnings advantage is greater. The coefficient on women's internal promotion ratios is not statistically significant. Together, it appears that men benefit from industries where they are able to leverage external offers and move between firms.

In Model 4, the explanatory variable is the difference in these ratios for men and women, with all controls now included. The coefficient for the difference in ratios for men and women seeks to capture how different the labor markets that men and women navigate are. This model shows that women's earnings are greater when more of men's earnings increases happen internal to the firm ($1,324$, $p < 0.001$). This model again supports the notion that men's earnings pull away from women's when they can access external labor markets. In industries where this option is less relevant, the earnings gap remains smaller.

**Robustness to Industry and Quarter-Year Fixed Effects**

Table 4 repeats the analyses of Tables 2 in the same sequence but introduces industry fixed effects alongside quarter-year fixed effects. When industry fixed effects are included, the model becomes a test of the association between a within-industry *change* in hiring and the earnings gap. Thus, in

| Residual earnings gap (women-men) | (1) | (2) | (3) | (4) |
|---|---|---|---|---|
| Prop. New | 6,094*** | | | |
| | (1,163) | | | |
| Ratio Int./Tot. Events | | 2,472*** | | |
| | | (425.2) | | |
| Ratio Int./Tot. Events, Men | | | 3,040*** | |
| | | | (416.3) | |
| Ratio Int./Tot. Events, Women | | | -328.8 | |
| | | | (204) | |
| $\Delta$ Ratio Men-Women | | | | 1324*** |
| | | | | (214.5) |
| Growth | -2564*** | -979.7*** | -1030*** | -415.4*** |
| | (397.5) | (168.1) | (169.5) | (137.9) |
| Advancement Speed | -70,310 | -20,420 | -17,740 | -32,760 |
| | (72,660) | (72,720) | (72,660) | (72,290) |
| Gender Composition | -12,530*** | -12,330*** | -12,390*** | -12,080*** |
| | (2,959) | (2,973) | (2,953) | (2,996) |
| Gender Composition, Squared | 7,683* | 7,589 | 7,570 | 7,565 |
| | (3,908) | (3,930) | (3,905) | (3,943) |
| Const. | -4,290 | -5,531 | -5,658 | -3,338 |
| | (609.8) | (765.1) | (763.4) | (541) |
| Quarter-Year FEs | ✓ | ✓ | ✓ | ✓ |
| $N$ | 102,000 | 102,000 | 102,000 | 102,000 |
| $R^2$ | 0.09221 | 0.0871 | 0.08998 | 0.08465 |

+ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 3: Association of residual gender earnings gap with proportion of internal earnings increase events, with controls

Note: Standard errors are clustered on industry. These are the 'adjusted' models including controls. In addition to the controls reported here, all models include quarter-year fixed effects. The dependent variable is the residual gender earnings gap. Model 1 shows the coefficient for the proportion of new workers (churn), Model 2 for the proportion of internal promotions of all earnings increase events. Model 3 shows this ratio separately for men and women. Model 4 shows the difference between these two ratios. In all models, an earnings increase is defined as an increase of at least 10% this quarter compared to last quarter, which is sustained (lasts for at least two quarters at the same employer). Controls are: "growth", the difference between this and last quarter's employment over the average of this and last year's employment; "Advancement Speed", the sum of all promotions (internal and external) over total workers; and "Gender Composition", the proportion of men in an industry-quarter-year. Standard errors are clustered on industry.

this table the models show gender earnings gaps when churn and the ratio of firm-internal to total earnings events increases, rather than testing the difference in levels.

In Model 1, when industry and quarter-year fixed effects are included, there is no longer a statistically significant coefficient on the proportion of new workers. Indeed, the sign flips compared to the models shown in Tables 2 and 3, which, if significant, would suggest that the gender earnings gap widens when churn increases in an industry.

In Model 2 of Table 4, the residual earnings gap is regressed on the ratio of internal promotions to all earnings increase events. Based on these results, we would expect to see an approximately a \$106 per year decline in the gender earnings gaps when an industry increases its ratio of internal promotions by one standard deviation up.

In Model 3, I turn again to examine the association of the residual gender earnings gap on the internal promotion ratio separately for men and for women, now with the industry and quarter-year fixed effects. The coefficient for men's ratio is 867.7 ($p < 0.001$), from which we can infer that there is \$195 decline in the gender earnings gap as men's ratio of internal promotions goes up by a standard deviation. On the other hand, holding men's earnings increase events constant, women in an industry increasing their share of internal promotions over all earnings increase events by a standard deviation would be associated with a \$83 *widening* of the gender earnings gap.

In Model 4, the explanatory variable is the gender difference in the ratio of internal promotions over all earnings increase events. The coefficient is positive (532.1, $p < 0.001$), which means that as more of men's earnings increase events are firm-internal as opposed to women's, the gender pay gap narrows. A one standard deviation shift up in this ratio would be associated with a \$92 closing of the gender earnings gap. Moving from being a $5th$ percentile to a $95th$ percentile industry would be associated with the gender gap closing in that industry by \$263.

The main findings are thus robust to the inclusion of industry and quarter-year fixed effects, though coefficients are markedly attenuated.

In sum, these analyses support the hypothesis that internal labor market reliance protects women's earnings. The findings regarding churn, measured as the proportion of new hires, are

difficult to fully square with the hypothesis. The main finding is that a narrower residual gender earnings gap is associated with more earnings increase events occuring within firms. This finding is robust across specifications (see Appendix A for additional tests). Further, the results of the models where gender-differentiated explanatory variables are included show that the more of *men's* earnings increase events happen within the firm, the narrower the gender earnings gap. This lends tentative support to the mechanisms described as driving the hypothesis: that men are better able to leverage the presence of an external labor market to improve their earnings.

| Residual earnings gap (women-men) | (1) | (2) | (3) | (4) |
|---|---|---|---|---|
| Prop. New | -531.9 | | | |
| | (339.3) | | | |
| Ratio Int./Tot. Events | | 496.8*** | | |
| | | (164.1) | | |
| Ratio Int./Tot. Events, Men | | | 867.7*** | |
| | | | (144) | |
| Ratio Int./Tot. Events, Women | | | -349*** | |
| | | | (86.39) | |
| $\Delta$ Ratio Men-Women | | | | 532.1*** |
| | | | | (81.3) |
| Const. | -3,762 | -4,438 | -4,465 | -4,005 |
| | (240.7) | (269.1) | (254) | (220.3) |
| Quarter-Year FEs | ✓ | ✓ | ✓ | ✓ |
| Industry FEs | ✓ | ✓ | ✓ | ✓ |
| $N$ | 102,000 | 102,000 | 102,000 | 102,000 |
| $R^2$ | 0.6108 | 0.6109 | 0.6112 | 0.6111 |

+ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 4: Association of residual gender earnings gap with proportion of internal earnings increase events, with industry fixed effects

Note: Standard errors are clustered on industry. Each model includes fixed effects for industry and quarter-year. The dependent variable is the residual gender earnings gap. Model 1 shows the coefficient for the proportion of new workers (churn), Model 2 for the proportion of internal promotions of all earnings increase events. Model 3 shows this ratio separately for men and women. Model 4 shows the difference between these two ratios. In all models, an earnings increase is defined as an increase of at least 10% this quarter compared to last quarter, which is sustained (lasts for at least two quarters at the same employer).

# Discussion

I hypothesized that the residual earnings gap between men and women is greater in labor markets with greater reliance on external hiring. In labor markets where internal promotions are the more dominant pathway to earnings advancement, women bridge the earnings gap. In this chapter, I used high quality commingled administrative data from the US Census Bureau and Internal Revenue Service to test this hypothesis. Across a series of model specifications, I find robust support for my hypothesis. My analyses show a strong relationship between greater unexplained earnings for men and a greater proportion of earnings increases – particularly of *men's* earnings increases – happening from moves between firms. It seems, then, that the more promotions occur internal to a firm, the better protected women are from discrimination, while an increased reliance on external labor markets may contribute to the stalled closing of the gender pay gap.

I have not, in this chapter, tested specific mechanisms for *why* external labor markets are associated with a greater residual gender earnings gap. I have theorized, standing on the shoulders of scholars who have preceded me, that bias and discrimination look different when evaluating a known employee than when evaluating a prospective outsider. I have also outlined the differences in legal protections for these two types of career junctures. The associational evidence I provide lends support for these theories, insomuch as such evidence can.

Some research has already looked at cross-firm effects of firm-internal changes, e.g. Wilmers (2019). I suggest future research into gender and the labor market also move to considering industries or local labor markets as a whole. If labor markets are to move to increased external hiring more generally, and many secular trends would have us expect this (Bidwell et al., 2013), it would be worthwhile to understand how minority workers can be supported in this new labor market landscape.

Further, there is no reason to think the patterns I observe are limited to the higher earning strata of the labor market. In fact, Andersson et al. (2005) find that the ability to change firms is most important for low-wage workers for whom changing firms can be the only way to escape working

poverty. Earnings inequality is widening between the bottom and top deciles of women more strongly than it is for men (Gharehgozli and Atal, 2020). Thus, the hurdles posed by the hiring interface demands the attention of those interested in understanding – and combating – widening inequality and deepening poverty.

The results presented in this chapter suggest that women might incur earnings increases were they to successfully leverage between-firm competition for talent. The theories I have discussed present a world of bias and discrimination, one where a woman can count herself lucky to have gained entry to any firm, with no hopes of securing another. This is not quite right or complete. Women do not apply to jobs as broadly as men do (see Chapter 3 of this dissertation), they work in occupations that may not allow for as much mobility (Benson, 2014), and their attempts to change jobs may not be taken seriously by employers (Rivera, 2017). For whatever reason, women may be less likely to apply to new jobs or pursue new leads to use purely as leverage at their current employer (though see Leslie et al. (2017) for exceptional women). Future research could expand our understanding of the barriers to firm-change and suggest policy interventions to produce more equitable outcomes. My analyses rely on industry-level variation in the data. The presence of this variation suggests future research could delve into specific industries to understand why, in some, women's earnings might increase as a result of cross-firm moves, while in others not. Potential avenues include looking at variation by the gender composition of industries, or by the level of earnings inequality as a whole in the industry. Most likely the story will be a complex one, but one worth disentangling.

# References

ANDERSSON, F., H. J. HOLZER, AND J. I. LANE (2005): *Moving Up or Moving On: Who Gets Ahead in the Low-Wage Labor Market?*, Russell Sage Foundation.

ANGEL, S., F. DISSLBACHER, S. HUMER, AND M. SCHNETZER (2019): "What did you really

earn last year?: explaining measurement error in survey income data," *Journal of the Royal Statistical Society: Series A (Statistics in Society)*, 182, 1411–1437.

AUTOR, D. H., L. F. KATZ, AND M. S. KEARNEY (2008): "Trends in US wage inequality: Revising the revisionists," *The Review of Economics and Statistics*, 90, 300–323.

AVENT-HOLT, D. AND D. TOMASKOVIC-DEVEY (2019): "Organizations as the Building Blocks of Social Inequalities," *Sociology Compass*, 13, e12655.

BALDI, S. AND D. B. MCBRIER (1997): "Do the Determinants of Promotion Differ for Blacks and Whites? Evidence from the US Labor Market," *Work and Occupations*, 24, 478–497.

BARNETT, W. P., J. N. BARON, AND T. E. STUART (2000): "Avenues of Attainment: Occupational Demography and Organizational Careers in the California Civil Service," *American Journal of Sociology*, 106, 88–144.

BENSON, A. (2014): "Rethinking the two-body problem: The segregation of women into geographically dispersed occupations," *Demography*, 51, 1619–1639.

BERTHEAU, A. (2021): "Employer Search Behavior: Reasons for Internal Hiring," *Labour Economics*, 73, 102064.

BERTRAND, M., C. GOLDIN, AND L. F. KATZ (2010): "Dynamics of the gender gap for young professionals in the financial and corporate sectors," *American Economic Journal: Applied Economics*, 2, 228–55.

BERTRAND, M., E. KAMENICA, AND J. PAN (2015): "Gender identity and relative income within households," *The Quarterly Journal of Economics*, 130, 571–614.

BIDWELL, M. (2011): "Paying more to get less: The effects of external hiring versus internal mobility," *Administrative Science Quarterly*, 56, 369–407.

BIDWELL, M., F. BRISCOE, I. FERNANDEZ-MATEO, AND A. STERLING (2013): "The Employment Relationship and Inequality: How and Why Changes in Employment Practices are Reshaping Rewards in Organizations," *Academy of Management Annals*, 7, 61–121.

BIDWELL, M. AND E. MOLLICK (2015): "Shifts and ladders: Comparing the role of internal and external mobility in managerial careers," *Organization Science*, 26, 1629–1645.

BIDWELL, M., S. WON, R. BARBULESCU, AND E. MOLLICK (2015): "I used to work at Goldman Sachs! How firms benefit from organizational status in the market for human capital," *Strategic Management Journal*, 36, 1164–1173.

BLAU, F. D. AND A. H. BELLER (1988): "Trends in earnings differentials by gender, 1971–1981," *ILR Review*, 41, 513–529.

BLAU, F. D., P. BRUMMUND, AND A. Y.-H. LIU (2013): "Trends in occupational segregation by gender 1970–2009: Adjusting for the impact of changes in the occupational coding system," *Demography*, 50, 471–492.

BLAU, F. D. AND L. M. KAHN (2008): "Women's work and wages," *The New Palgrave Dictionary of Economics*, 8, 762–72.

——— (2017): "The gender wage gap: Extent, trends, and explanations," *Journal of Economic Literature*, 55, 789–865.

BOUND, J. AND A. B. KRUEGER (1991): "The extent of measurement error in longitudinal earnings data: Do two wrongs make a right?" *Journal of labor Economics*, 9, 1–24.

BRETT, J. M. AND L. K. STROH (1997): "Jumping Ship: Who Benefits from an External Labor Market Career Strategy?" *Journal of Applied Psychology*, 82, 331.

BRUNS, B. (2019): "Changes in workplace heterogeneity and how they widen the gender wage gap," *American Economic Journal: Applied Economics*, 11, 74–113.

BUCHMANN, C. AND T. A. DIPRETE (2006): "The growing female advantage in college completion: The role of family background and academic achievement," *American Sociological Review*, 71, 515–541.

BUCHMANN, C., T. A. DIPRETE, AND A. MCDANIEL (2008): "Gender inequalities in education," *Annual Review of Sociology*, 34, 319–337.

BUDIG, M. J. (2002): "Male advantage and the gender composition of jobs: Who rides the glass escalator?" *Social Problems*, 49, 258–277.

BUSER, T., M. NIEDERLE, AND H. OOSTERBEEK (2014): "Gender, competitiveness, and career choices," *The Quarterly Journal of Economics*, 129, 1409–1447.

CAMPBELL, B. A., R. COFF, AND D. KRYSCYNSKI (2012): "Rethinking sustained competitive advantage from human capital," *Academy of Management Review*, 37, 376–395.

CARD, D., A. R. CARDOSO, J. HEINING, AND P. KLINE (2018): "Firms and labor market inequality: Evidence and some theory," *Journal of Labor Economics*, 36, S13–S70.

CARD, D., A. R. CARDOSO, AND P. KLINE (2016): "Bargaining, sorting, and the gender wage gap: Quantifying the impact of firms on the relative pay of women," *The Quarterly Journal of Economics*, 131, 633–686.

CARD, D., J. HEINING, AND P. KLINE (2013): "Workplace Heterogeneity and the Rise of West German Wage Inequality," *The Quarterly Journal of Economics*, 128, 967–1015.

CASTILLA, E. J. (2008): "Gender, Race, and Meritocracy in Organizational Careers," *American Journal of Sociology*, 113, 1479–1526.

CECI, S. J., D. K. GINTHER, S. KAHN, AND W. M. WILLIAMS (2014): "Women in academic science: A changing landscape," *Psychological Science in the Public Interest*, 15, 75–141.

CIMPIAN, J. R., T. H. KIM, AND Z. T. MCDERMOTT (2020): "Understanding persistent gender gaps in STEM," *Science*, 368, 1317–1319.

CORRELL, S. J., S. BENARD, AND I. PAIK (2007): "Getting a job: Is there a motherhood penalty?" *American Journal of Sociology*, 112, 1297–1338.

CRUZ, G. AND T. RAU (2022): "The effects of equal pay laws on firm pay premiums: Evidence from Chile," *Labour Economics*, 75, 102135.

DEVARO, J., A. KAUHANEN, AND N. VALMARI (2019): "Internal and external hiring," *ILR Review*, 72, 981–1008.

DREHER, G. F. AND T. H. COX JR (2000): "Labor Market Mobility and Cash Compensation: The Moderating Effects of Race and Gender," *Academy of Management Journal*, 43, 890–900.

ENGLAND, P. (2010): "The gender revolution: Uneven and stalled," *Gender & Society*, 24, 149–166.

FULLER, S. (2008): "Job Mobility and Wage Trajectories for Men and Women in the United States," *American Sociological Review*, 73, 158–183.

GHAREHGOZLI, O. AND V. ATAL (2020): "Revisiting the gender wage gap in the United States," *Economic Analysis and Policy*, 66, 207–216.

GILLEY, K. M. AND A. RASHEED (2000): "Making more by doing less: an analysis of outsourcing and its effects on firm performance," *Journal of Management*, 26, 763–790.

GITTLEMAN, M. (2022): "The "Great Resignation" in perspective," *Monthly Labor Review, U.S. Bureau of Labor Statistics*, July.

GOLDIN, C. (2006): "The quiet revolution that transformed women's employment, education, and family," *American Economic Review*, 96, 1–21.

——— (2014): "A grand gender convergence: Its last chapter," *American Economic Review*, 104, 1091–1119.

GOLDIN, C. AND C. ROUSE (2000): "Orchestrating impartiality: The impact of" blind" auditions on female musicians," *American Economic Review*, 90, 715–741.

GOLDSCHMIDT, D. AND J. F. SCHMIEDER (2017): "The rise of domestic outsourcing and the evolution of the German wage structure," *The Quarterly Journal of Economics*, 132, 1165–1217.

GOUGH, M. AND M. NOONAN (2013): "A review of the motherhood wage penalty in the United States," *Sociology Compass*, 7, 328–342.

HART, C. G. (2021): "Trajectory guarding: Managing unwanted, ambiguously sexual interactions at work," *American Sociological Review*, 86, 256–278.

HECKMAN, J. J. AND T. KAUTZ (2012): "Hard evidence on soft skills," *Labour Economics*, 19, 451–464.

HJORT, J., X. LI, AND H. SARSONS (2019): "Across-Country Wage Compression in Multinationals," Tech. rep., mimeo, Columbia University.

HYATT, H. R. AND J. R. SPLETZER (2013): "The recent decline in employment dynamics," *IZA Journal of Labor Economics*, 2, 1–21.

JONES, D. R. AND G. H. MAKEPEACE (1996): "Equal worth, equal opportunities: pay and promotion in an internal labour market," *The Economic Journal*, 106, 401–409.

KALEV, A. AND F. DOBBIN (2006): "Enforcement of Civil Rights Law in Private Workplaces: The Effects of Compliance Reviews and Lawsuits over Time," *Law & Social Inquiry*, 31, 855–903.

KALEV, A., F. DOBBIN, AND E. KELLY (2006): "Best Practices or Best Guesses? Assessing the Efficacy of Corporate Affirmative Action and Diversity Policies," *American Sociological Review*, 71, 589–617.

KELLER, J. (2018): "Posting and slotting: How hiring processes shape the quality of hire and compensation in internal labor markets," *Administrative Science Quarterly*, 63, 848–878.

KOCHHAR, R., K. PARKER, AND R. IGIELNIK (2022): "Majority of U.S Workers Changing Jobs Are Seeing Real Wage Gains," *Pew Research Center*, July.

KRONBERG, A.-K. (2013): "Stay or leave? Externalization of job mobility and the effect on the US gender earnings gap, 1979-2009," *Social Forces*, 91, 1117–1146.

LAZEAR, E. P. AND J. R. SPLETZER (2012): "Hiring, churn, and the business cycle," *American Economic Review*, 102, 575–79.

LEONARDI, M., M. PELLIZZARI, AND D. TABASSO (2019): "Wage compression within the firm: Evidence from an indexation scheme," *The Economic Journal*, 129, 3256–3291.

LESLIE, L. M., C. F. MANCHESTER, AND P. C. DAHM (2017): "Why and when does the gender gap reverse? Diversity goals and the pay premium for high potential women," *Academy of Management Journal*, 60, 402–432.

LEUNG, M. D. (2014): "Dilettante or renaissance person? How the order of job experiences affects hiring in an external labor market," *American Sociological Review*, 79, 136–158.

LEVANON, A., P. ENGLAND, AND P. ALLISON (2009): "Occupational feminization and pay: Assessing causal dynamics using 1950–2000 US census data," *Social Forces*, 88, 865–891.

LI, J., B. DOSTIE, AND G. SIMARD-DUPLAIN (2022): "Firm Pay Policies and the Gender Earnings Gap: The Mediating Role of Marital and Family Status," *ILR Review*, 00197939221093562.

LUNDBERG, S. AND J. STEARNS (2019): "Women in economics: Stalled progress," *Journal of Economic Perspectives*, 33, 3–22.

MOSS-RACUSIN, C. A., J. F. DOVIDIO, V. L. BRESCOLL, M. J. GRAHAM, AND J. HANDELSMAN (2012): "Science faculty's subtle gender biases favor male students," *Proceedings of the National Academy of Sciences*, 109, 16474–16479.

MULLIGAN, C. B. AND Y. RUBINSTEIN (2008): "Selection, investment, and women's relative wages over time," *The Quarterly Journal of Economics*, 123, 1061–1110.

NEUMARK, D., R. J. BANK, AND K. D. VAN NORT (1996): "Sex discrimination in restaurant hiring: An audit study," *The Quarterly Journal of Economics*, 111, 915–941.

NGUYEN, H., B. R. PARKER, AND S. S. SIMPSON (2022): "Equal Pay for Equal Work? Considering the Gender Gap in Illegal Pay," *Journal of Quantitative Criminology*, 38, 425–458.

NIEDERLE, M. AND L. VESTERLUND (2007): "Do women shy away from competition? Do men compete too much?" *The Quarterly Journal of Economics*, 122, 1067–1101.

PARKER, K. AND J. MENASCE HOROWITZ (2022): "Majority of workers who quit a job in 2021 cite low pay, no opportunities for advancement, feeling disrespected," *Pew Research Center*, July.

PEDULLA, D. S. (2016): "Penalized or Protected? Gender and the consequences of Nonstandard and Mismatched Employment Histories," *American Sociological Review*, 81, 262–289.

PEKKARINEN, T. AND J. VARTIAINEN (2006): "Gender differences in promotion on a job ladder: Evidence from Finnish metalworkers," *ILR Review*, 59, 285–301.

PETERSEN, T. AND I. SAPORTA (2004): "The Opportunity Structure for Discrimination," *American Journal of Sociology*, 109, 852–901.

QUADLIN, N. (2018): "The mark of a woman's record: Gender and academic performance in hiring," *American Sociological Review*, 83, 331–360.

QUINTANA-GARCIA, C. AND M. M. ELVIRA (2017): "The effect of the external labor market on the gender pay gap among executives," *ILR Review*, 70, 132–159.

RIVERA, L. A. (2017): "When two bodies are (not) a problem: Gender and relationship status discrimination in academic hiring," *American Sociological Review*, 82, 1111–1138.

SHAUMAN, K. A. (2010): "Gender asymmetry in family migration: Occupational inequality or interspousal comparative advantage?" *Journal of Marriage and Family*, 72, 375–392.

SHAUMAN, K. A. AND Y. XIE (1996): "Geographic mobility of scientists: Sex differences and family constraints," *Demography*, 33, 455–468.

SONG, J., D. J. PRICE, F. GUVENEN, N. BLOOM, AND T. VON WACHTER (2019): "Firming up inequality," *The Quarterly Journal of Economics*, 134, 1–50.

SORKIN, I. (2017): "The role of firms in gender earnings inequality: Evidence from the United States," *American Economic Review*, 107, 384–87.

WANG, H. C. AND J. B. BARNEY (2006): "Employee incentives to make firm-specific investments: Implications for resource-based theories of corporate diversification," *Academy of Management Review*, 31, 466–476.

WILLIAMS, C. L. (1992): "The glass escalator: Hidden advantages for men in the "female" professions," *Social Problems*, 39, 253–267.

WILLIAMS, W. M. AND S. J. CECI (2015): "National hiring experiments reveal 2: 1 faculty preference for women on STEM tenure track," *Proceedings of the National Academy of Sciences*, 112, 5360–5365.

WILMERS, N. (2019): "Solidarity within and across workplaces: How cross-workplace coordination affects earnings inequality," *RSF: The Russell Sage Foundation Journal of the Social Sciences*, 5, 190–215.

WILMERS, N. AND W. KIMBALL (2022): "How Internal Hiring Affects Occupational Stratification," *Social Forces*, 101, 111–149.

## Using a stricter definition of earnings increase event

Tables 5, 6, and 7 replicate the analyses of Tables 2, 3, and 4, respectively, this time using a stricter definition of "earnings increase event". Here, an earnings increase only counts as such if it constitutes a minimum 20 percent increase from the previous quarter's earnings and remains at this level or higher into the next quarter. All findings are replicated using this stricter definition.

| Residual earnings gap (Women-men) | (1) | (2) | (3) | (4) |
|---|---|---|---|---|
| Prop. New | 427 | | | |
| | (514.7) | | | |
| Ratio Int./Tot. Events | | 358* | | |
| | | (181.8) | | |
| Ratio Int./Tot. Events, Men | | | 1962*** | |
| | | | (294.1) | |
| Ratio Int./Tot. Events, Women | | | -1444*** | |
| | | | (221) | |
| $\Delta$ Ratio Men-Women | | | | 1636*** |
| | | | | (233.7) |
| Const. | -5,958 | -6,090 | -6,194 | -5,893 |
| | (159.3) | (156.5) | (162.8) | (127.9) |
| $N$ | 102,000 | 102,000 | 102,000 | 102,000 |
| $R^2$ | 0.0001067 | 0.0001976 | 0.004330 | 0.0039150 |

Table 5: Residual gender earnings gap on earnings increases, 20% threshold

Note: Standard errors are clustered on industry. These are the 'unadjusted' models – there are no further controls not reported here. The dependent variable is the residual gender earnings gap. Model 1 shows the coefficient for the proportion of new workers (churn), Model 2 for the proportion of internal promotions of all earnings increase events. Model 3 shows this ratio separately for men and women. Model 4 shows the difference between these two ratios. In all models, an earnings increase event is defined as an increase of at least 20% this quarter compared to last quarter, which is sustained (lasts for at least two quarters at the same employer).

| Residual earnings gap (Women-men) | (1) | (2) | (3) | (4) |
|---|---|---|---|---|
| Prop. New | 6,162*** | | | |
| | (1,167) | | | |
| Ratio Int./Tot. Events | | 1,682*** | | |
| | | (354.4) | | |
| Ratio Int./Tot. Events, Men | | | 2,891*** | |
| | | | (410.1) | |
| Ratio Int./Tot. Events, Women | | | -821.8*** | |
| | | | (199) | |
| $\Delta$ Ratio Men-Women | | | | 1,558*** |
| | | | | (228.7) |
| Growth | -2548*** | -707.8*** | -763.8*** | -393.5** |
| | (396.4) | (158.5) | (160.3) | (140.6) |
| Advancement Speed | -120,900 | -36,070 | -37,290 | -39,230 |
| | (85,700) | (84,650) | (84,610) | (84,790) |
| Gender Composition | -12,540*** | -12,290*** | -12,490*** | -12,130*** |
| | (2,957) | (2,979) | (2,943) | (2,986) |
| Gender Composition, Squared | 7,684* | 7,692 | 7,828* | 7,642 |
| | (3,907) | (3,932) | (3,885) | (3,928) |
| Const. | -4,257 | -4,797 | -5,100 | -3,346 |
| | (607.5) | (711.8) | (722.3) | (539.5) |
| Quarter-Year FEs | ✓ | ✓ | ✓ | ✓ |
| $N$ | 102,000 | 102,000 | 102,000 | 102,000 |
| $R^2$ | 0.0923 | 0.0848 | 0.0896 | 0.0862 |

Table 6: Residual gender earnings gap on earnings increases, with controls, 20% threshold

Note: Standard errors are clustered on industry. These are the 'adjusted' models including controls. In addition to the controls reported here, all models include quarter-year fixed effects. The dependent variable is the residual gender earnings gap. Model 1 shows the coefficient for the proportion of new workers (churn), Model 2 for the proportion of internal promotions of all earnings increase events. Model 3 shows this ratio separately for men and women. Model 4 shows the difference between these two ratios. In all models, an earnings increase event is defined as an increase of at least 20% this quarter compared to last quarter, which is sustained (lasts for at least two quarters at the same employer). Controls are: "growth", the difference between this and last quarter's employment over the average of this and last year's employment; "Advancement Speed", the sum of all earnings increase events (internal and external) over total workers; and "Gender Composition", the proportion of men in an industry-quarter-year.

| | (1) | (2) | (3) | (4) |
|---|---|---|---|---|
| Prop. New | -531.9 | | | |
| | (339.3) | | | |
| Ratio Int./Tot. Events | | 573.6*** | | |
| | | (149.9) | | |
| Ratio Int./Tot. Events, Men | | | 945.7*** | |
| | | | (130.2) | |
| Ratio Int./Tot. Events, Women | | | -341.1*** | |
| | | | (77.35) | |
| $\Delta$ Ratio Men-Women | | | | 554.7*** |
| | | | | (73) |
| Const. | -3,762 | -4,509 | -4,548 | -4,038 |
| | (240.7) | (263.3) | (245.8) | (219.8) |
| Quarter-Year FEs | ✓ | ✓ | ✓ | ✓ |
| Industry FEs | ✓ | ✓ | ✓ | ✓ |
| $N$ | 102,000 | 102,000 | 102,000 | 102,000 |
| $R^2$ | 0.6108 | 0.6110 | 0.6114 | 0.6112 |

Table 7: Residual gender earnings gap on earnings increases, with industry fixed effects, 20% threshold

Note: Standard errors are clustered on industry. Each model includes fixed effects for industry and quarter-year. The dependent variable is the residual gender earnings gap. Model 1 shows the coefficient for the proportion of new workers (churn), Model 2 for the proportion of internal promotions of all earnings increase events. Model 3 shows this ratio separately for men and women. Model 4 shows the difference between these two ratios. In all models, an earnings increase event is defined as an increase of at least 20% this quarter compared to last quarter, which is sustained (lasts for at least two quarters at the same employer).